



Prostate cancer grade using self-supervised learning and novel feature aggregator based on weakly-labeled gbit-pixel pathology images

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Abstract

Prostate cancer (PCa) is the second most common cancer in men worldwide. The Gleason score, determined by pathologists through microscopic examination of pathological tissue, is the most powerful prognostic indicator for patients diagnosed with prostate cancer. However, there is inter- and intra-observer variability among pathologists, and the scoring process imposes a significant workload on them. This study presents a deep learning model that utilizes gigapixel pathology images and slide-level labels for prostate cancer detection and Gleason grading. WSIs are first cropped into small patches, and a deep learning model trained using self-supervised learning is used to extract features from the patches. An attention-LSTM is then used to aggregate features and perform the final classification. Our pipeline for Gleason grading, which utilizes a 6-level classification system, achieves a quadratic weighted kappa of 0.930 on the internal test dataset ($n = 531$). Furthermore, it demonstrates good generalization on the external test dataset ($n = 155$) with a quadratic weighted kappa of 0.8668. The proposed model integrating a self-supervised feature extraction model and an attention LSTM aggregator, is one of the most advanced scoring methods for grading prostate cancer using only slide-level labels. It represents a significant advancement in automating the extraction of prostate cancer grading information in digital diagnostic pathology and will help to address the critical shortage of pathologists.

Keywords Prostate cancer grade · Deep convolutional network · Self-supervised learning · Attention-LSTM

1 Introduction

Prostate cancer is the most prevalent cancer among men in the United States, with approximately 248,500 new cases reported in 2021, accounting for 26% of all new cancer cases in men. It is also the second leading cause of cancer death among men in the United States, with an estimated 34,100 deaths in 2021. This accounts for 11% of all cancer deaths in men [1]. Most men diagnosed with prostate cancer will not die quickly, and early detection and overtreatment may

not improve patients' life expectancy. Therefore, it is important to assess the aggressiveness of prostate cancer and treat it accordingly [2]. To assess the aggressiveness of prostate cancer, pathologists often use the Gleason system, which classifies prostate cancer into five grades ranging from 1 (low risk) to 5 (high risk) based on the pathological biopsy. The Gleason score in biopsies is determined by adding the most common pattern to the highest secondary pattern (e.g., 3+5). In recent years, the Gleason system has undergone some modifications since its inception. It now defines Gleason scores of 3 + 3 = 6 as Group 1, scores of 3 + 4 = 7 as Group 2, scores of 4 + 3 = 7 as Group 3, scores of 8 as Group 4, and scores of 9 and 10 as Group 5 [3].

As the number of prostate cancer cases increases, it becomes increasingly challenging to rely solely on a limited workforce to review pathology images for Gleason grading. Furthermore, the grading patterns comprising the grades and groups (mainly grades 3 and 4), are still susceptible to inter- and intra-observer variability. One solution to this challenge is the development of an artificial intelligence (AI)

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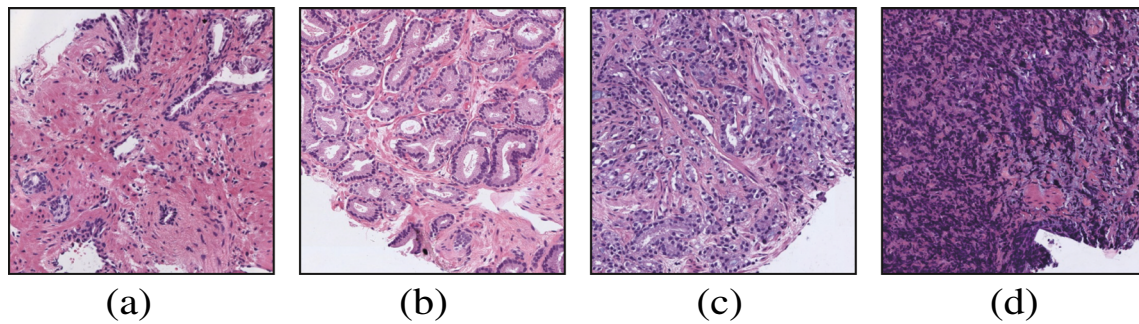


Fig. 1 (a) region containing benign glands. (b) region containing GG3 glandular structures. (c) region containing GG4 patterns. (d) region containing GG5 patterns

CAD system for the automated grading of prostate pathology images. However, current deep learning-based CAD systems for processing pathological images can encounter issues. Firstly, it is not possible to directly train end-to-end with pathological whole slide images (WSI), which can be as large as 10,000 × 10,000 pixels [4]. Secondly, the proportion of diseased tissue is small, making it difficult to identify the key tissue. In this paper, a weakly supervised Gleason grading pipeline was proposed beginning by cropping whole slide images (WSI) into patches. Then, embeddings of these patches were obtained using a feature extraction model that had been pre-trained through self-supervised learning on the TCGA dataset [5]. Finally, a novel feature aggregator was designed to diversely aggregate information contained in patch collections in order to generate slide-level labels (Fig. 1).

2 Related work

Traditional CAD-based Gleason scoring systems typically design hand-crafted features on the basis of domain knowledge and employ standard classifiers, such as support vector machines (SVMs) [6], random forests [7] or Bayesian classifiers [8] to perform classification. Recently, deep learning technology has shown promise for diagnosis in several medical fields, including radiology, ophthalmology, dermatology, and pathology [9]. In histopathology, deep learning (DL) methods require pixel-level or patch-level annotations to prevent memory bottlenecks caused by the gigapixel size of whole slide images (WSIs). For example, Lucas et al. [10] were early pioneers in applying the DL method for Gleason scoring which developed an Inception v3 architecture to discriminate between Gleason grade 3 and higher patterns based on 96 pixel-level annotated prostate biopsies from 38 patients. Otalora et al. [11] proposed a two-stage deep learning method, in which the CNN model was initially trained on regionally annotated patches and subsequently fine-tuned using image-level labels. Their methods demonstrated that

fine-tuning at the slide level has a substantial effect on the performance of CNN models for the Gleason scoring task. Silva-Rodríguez et al. [12] trained a three-layer CNN model from scratch for patch-level Gleason grading. They also carried out innovative work on the detection of cribriform patterns and achieved a high AUC of 0.82. Nagpal et al. [13] trained a region-level CNN based on pathologists' annotations. During training, each partially trained model was periodically applied to the entire training corpus of whole slide images in order to conduct hard negative mining. Finally, the k-nearest neighbor (kNN) algorithm was trained to predict GG by calibrating and transforming the outputs of the region-based CNN into four features that indicate the extent of tumor presence on the slide. Similar work has been done by Arvaniti et al. [14] and Bulten et al. [15]. Although these methods achieved high performance, they required additional patch-level or pixel-level annotation from the pathologists. Furthermore, an additional step was necessary to obtain the slide-level labels.

Compared to patch-level supervised methods, slide-level supervised CAD systems can be more popular due to their less labor-intensive labeling property, where labels can be derived directly from pathology reports. Yechan et al. [18] presented a novel deep learning-based Gleason grading system that does not require extensive expert region-level annotation. Instead, it automatically generates region-level annotations on a total of 6,664 and 936 prostate needle biopsy slides (689 and 99 cases) from two institutions. Hammouda et al. [19] converted the Gleason pattern recognition problem into a multilevel binary classification problem based on multi-resolution patches and demonstrated that multi-level binary classification can effectively strengthen the model's scoring ability. Wenyan et al. [20] proposed a novel joint task using Mask-RCNN for epithelial cell detection and CNN for Gleason grading and confirmed that both tasks can gain performance improvement when joint training is performed. Karimi et al. [8] presented a new method for combining information from patches of different sizes, as well as a novel data augmentation method that augments patches in the space

Table 1 Summary of the characteristics of studies reporting on Artificial Intelligence (AI) for prostate cancer pathological diagnosis

Authors	Datasets	Approaches	Metrics and performance
Lucas et al. [10]	96 tissue from 38 patients	Inception v3	Kappa = 0.70
Otálora et al. [11]	886 prostate from 641 patients 301 cases of WSIs	MobileNet	Kappa = 0.691 against pathologists
Arvaniti et al. [14]	641 patients and 245 patients	MobileNet	Kappa = 0.71
Silva-Rodríguez et al. [16]	Sicap [16]	three layer CNN	Kappa = 0.77 AUC, Kappa
Nagpal et al. [13]	1226 slides for training, 331 slides for validation	Inception v3 for patches and KNN for slide	Accuracy = 0.70
Silva-Rodríguez et al. [12]	Panda [17], Sicap [16]	MIL, pscocode training for patch-level GS	Kappa = 0.7954
Yechan et al. [18]	1133 prostate biopsy for train, 700 for validation	DeepLab v3+ and rule-based judgment	Kappa = 0.907
Hammouda et al. [19]	3080 WSIs of the digitized PBSs, 2888 for training, 96 for validation, and 77 for testing	Multilevel Binary Classification	Recall is 0.5 or 0.90, Precision is 0.5 or 0.92
Wenyuan et al. [20]	513 images retrieved from Cedars-Sinai Medical Center	Mask R-CNN	mIOU = 79.56, pixel-level accuracy = 0.894
Karimi et al. [8]	333 TMAs from 231 patients	multi-resolution CNNs and GAN	accuracy = 0.92 for cancer detection
Duran-Lopez et al. [21]	158 normal WSIs and 174 malignant WSIs	Wide and Deep CNN	Accuracy = 94.24%, Sensitivity = 98.87%
Bulten et al. [15]	5759 H&E-stained biopsies from 1243 patients	pathologists with AI assistance	kappa = 0.872
Jinxi et al. [22]	Panda [17], Sicap [16]	Resnet50 and GNN	AUC = 0.985, Kappa = 0.931

of learned features rather than the space of image patches. Their system had a Cohen's kappa score of 0.897 and a GG prediction accuracy of 77.5%. It's worth noting that these approaches described above were all trained on kbit-level pathology images. When the image size reaches Gbit, memory bottlenecks will be a hindrance to these methods. With regard to this, Pinckaers et al. [23] suggested the use of a streaming implementation of convolutional layers instead of traditional convolutions to train a modern CNN end-to-end to minimize GPU memory consumption. The algorithms presented in the PANDA competition proposed to slice WSI images into tiles and stitch informative tiles together to reduce the size of the image fed to the CNN model [17]. The results of the competition showed that this approach can achieve a high agreement with expert uropathologists (0.862 quadratic weighted kappa, 95% confidence interval) and save a significant amount of memory. The MIL paradigm was also an effective approach for handling weakly labeled pathological images. Campanella et al. [24] trained a ResNet34 model using the Multiple Instance Learning (MIL) paradigm, where the top-ranked tiles were selected to match the slide-level labels. During the inference process, an RNN was used to aggregate the tile information and generate the final slide-level labels. Silva-Rodríguez et al. [12] proposed to use the

patch with the highest predicted probability in the patch collection to perform loss back-propagation for the whole slide image. The trained model was then used to assign pseudo-labels to sets of patches. Finally, a student network was trained at the patch level to assign grades to the patches. Jiawen et al. [25] introduced the attention mechanism into the MIL paradigm to scale the embeddings of patches according to their attention scores. By utilizing the attention mechanism, it can aggregate the information encompassed in the most significant patches. The CLAM-MB [26] method was an improved version of MIL, which enhanced the discriminative power of each scoring classifier by constructing positive and negative patch samples and conducting binary classification training on the constructed samples (Table 1).

As deep learning techniques continue to evolve, advanced architectures such as EfficientNet, ViT, and Swin transformers are being increasingly utilized in the field of medical imaging [9, 27, 28]. The Visual Transformer (ViT) is an advanced architecture that originated in the field of Natural Language Processing (NLP). It is adept at capturing global information through the use of multi-headed attention. It can exist as a pure transformer or act as backbones, extracting features only, or as feature classifiers, aggregating features extracted by a CNN. When used as feature extractors, these

advanced architectures can be trained in a self-supervised manner. RetCCL [29] and CTransPath [30] are well-known Resnet50 and Vit backbones training with self-supervised learning on the TCGA pathology dataset. Jinxi et al. [22] proposed a scoring method based on the RetCCL backbone and GNN aggregator, which achieved high performance in cancer detection and Gleason grading tasks. It is worth noting that hyperspectral pathology imaging is also an area of great promise that contains richer dimensional information than RGB pathology [5, 31, 32]. 3D Convolutional neural networks, PCA mutual information dimensionality reduction, and Spectral Interval Convolution are some of the popular techniques used for processing hyperspectral images.

Despite the existence of these advanced methods, there is still room for improvement because little attention has been paid to feature selection during feature aggregation. In this paper, we utilized self-supervised learning-trained models as feature extractors to overcome memory limitations. Additionally, we introduced a novel attention-based feature aggregator to expand the diversity of the patch sequence and enhance the overall grading performance.

3 Materials and methods

3.1 Datasets

We used two public datasets to validate the effectiveness of our method. The first dataset is the PANDA dataset, which was proposed in the PANDA Challenge held in 2019 [17]. The dataset consists of 10,616 prostate whole slide images (WSI), in which the primary and secondary Gleason grades were marked by different pathologists. It has a test dataset containing 531 WSI images with unknown labels. The second dataset used is the SICAPv2 dataset containing 155 prostate WSIs [16]. The distribution of labels for these datasets can be seen in Table 2.

3.2 Image pre-processing

Cropping whole slide images (WSI) into patches of appropriate sizes is a necessary step in our pipeline. When performing the cropping, we utilize the patch generation step provided

Table 2 Biopsies-level data partition and Gleason Scores (GS) distribution of used datasets

Dataset	Gleason scores					
	GD6	GD7	GD8	GD9	GD10	Benign
PANDA	2666	1343	1242	1249	1166	2892
SICAPv2	14	22	23	18	1268	36

Table 3 The number of patches generated and the time taken using different strategies on the Panda dataset

Cropping size	Total patches	Cropping time
256x256	5574430	58h
512x512	1514588	36h
768x768	773703	20h

by the open-source tool CLAM-MB [26], which initially segments images (WSI) to exclude non-tissue regions. Image patches are then extracted from the tissue regions of the whole slide images (WSI) within the segmented foreground contours [33].

Our model performance is highly correlated with the cropping size and overlapping strategies. Some literature suggested that patches cropped to 512x512 at 10x resolution is the appropriate strategy for pathology classification tasks. Because this size is large enough to contain benign fusiform or dilated glands, as well as wedge-shaped GG4 structures, a smaller size would compromise the integrity of these structures. Smaller cropping sizes will also increase the number of patches generated for each WSI image and put a strain on the training resources of sequence-based aggregators. Patches with larger sizes won't work either because a significant amount of pixel information is discarded during the cropping and feature extraction process [12].

In terms of cropping and training efficiency, a size of 512x512 is more suitable for our pipeline. Table 3 shows the cropping time and the number of patches generated using various cropping strategies. From Table 3, we can see that cropping with a size of 256x256 results in three times the number of patches compared to the 512x512 strategy. This also implies three times the training time and feature extraction time. In summary, we decided to perform patch cropping at 20x magnification using a size of 512x512 for the PANDA dataset. For the SICAPv2 dataset, 155 whole slide images (WSI) were already cropped into 18,783 512x512 patches [16] (Fig. 2).

3.3 Feature extraction

The purpose of feature extraction is to learn a good representation of the learning objects. Methods such as Attention-MIL [34] and CLAM-MB [26] used ImageNet pre-trained weights in a supervised manner to extract the representation features. They perform poorly due to the fact that pathology images are significantly different from ImageNet images, and supervised learning cannot effectively utilize large amounts of unlabeled data. In this paper, we utilize contrastive learning to train feature extraction models. By making similar instances close together and dissimilar instances far part in

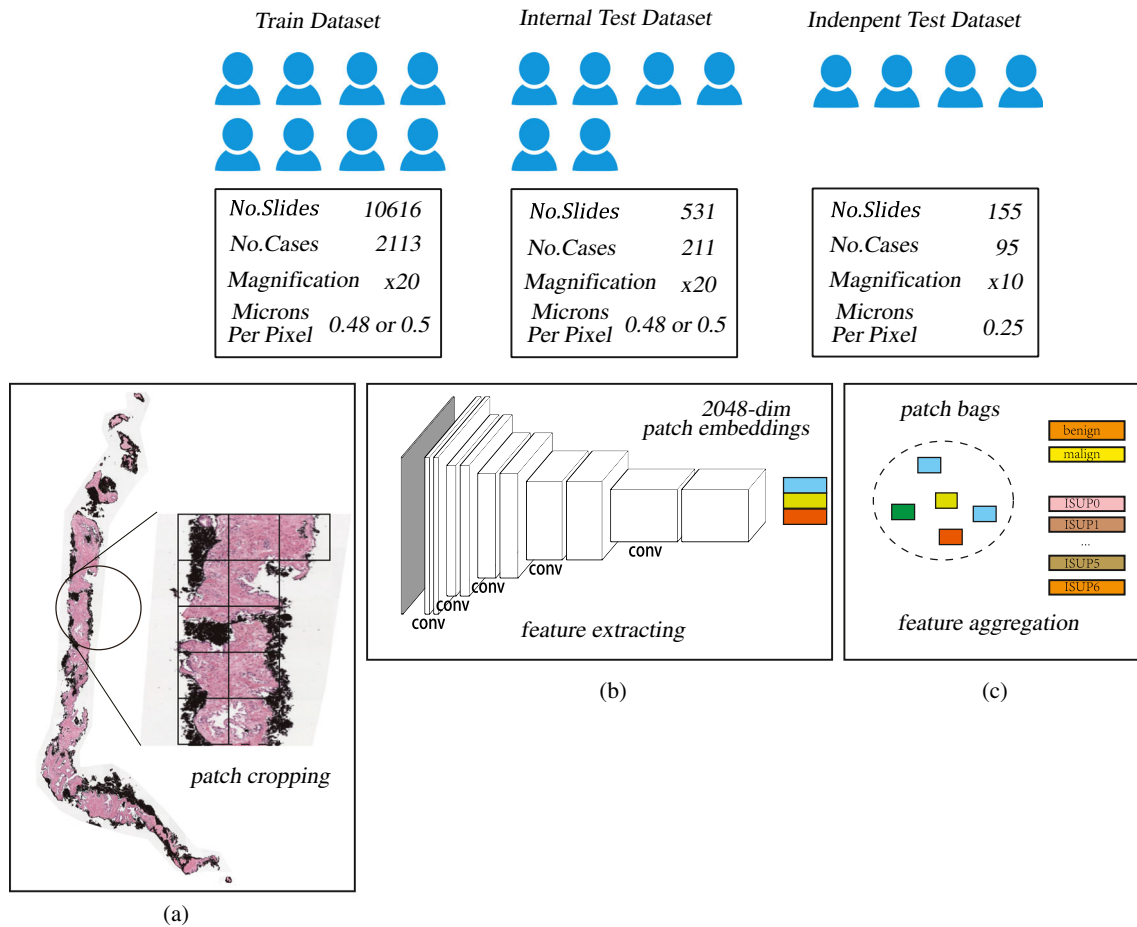


Fig. 2 The pipeline of our proposed method, there are three main steps in our weakly supervised attention LSTM method. (a) Cropping the whole slide images (WSI) into 512x512-sized patches; (b) extracting

the patch embeddings using a self-supervised pre-training resnet50; (c) aggregating the embeddings to obtain the slide level using a Bi-LSTM network with novel attention mechanism labels

the representation space, the feature extraction model can learn better representations of patches.

$$\mathcal{L} = \frac{1}{2N} \sum_{k=1}^N [\ell(2k-1, 2k) + \ell(2k, 2k-1)] \quad (1)$$

When training is done in a batch manner, we can get a good feature extractor by minimizing the loss function shown in (1) where N represents the batch size and ℓ indicates the loss function for a positive pair which can be defined as (2). In (2), z denotes the representation features extracted by the feature extractor. z_i and z_j are feature representations of similar images (usually different augmentation views of the same image) and z_k is feature representations extracted by images that are not similar to image i . sim is the similarity function which is usually defined as $\text{sim}(i, j) = z_i^\top z_j / (\|z_i\| \|z_j\|)$, $\Gamma_{k \neq i} \in \{0, 1\}$ is an indicator function that evaluates to 1 if $k \neq i$ and 0 if $k = i$. τ denotes a temperature parameter that

regulates the similarity function.

$$\ell_{i,j} = -\log \frac{\exp(\text{sim}(z_i, z_j) / \tau)}{\sum_{k=1}^{2N} \Gamma_{k \neq i} \exp(\text{sim}(z_i, z_k) / \tau)} \quad (2)$$

RETCCCL [5] trained a ResNet50 backbone using contrastive learning on the TCGA dataset, and we used these pre-trained weights for feature extraction. We resized the patches generated in the first step to half of their original size and fed them into the ResNet50 backbone model, following the instructions provided by RETCCCL. Finally we obtained feature vectors of length 2048 for all the patches, and these feature vectors were stored as h5 files for further supervised training.

3.4 Feature aggregation

After extracting all feature embeddings for patches, we need to aggregate these feature embeddings to obtain the Gleason

score or determine whether the pathology images are benign or not. Max-pooling is a simple but effective aggregator parameterized aggregators such as RNN, Attention-MIL, or GCN may be better choices due to their more powerful aggregation capabilities. The work of Bilal et al. [35] showed that the performance of diagnostic prediction can be improved with an appropriate patch selection strategy. Inspired by this, we proposed a novel feature aggregation method with dropout attention mechanisms to effectively increase the diversity of patch sequence samples.

$$L(\theta) = -\frac{1}{M} \sum_{i=1}^M \log p(y_i | f(x_i, z_i)) \quad (3)$$

$$\tilde{z} = \text{sigmoid} \left(\frac{1}{t} (\log \theta - \log(1 - \theta) + \log u - \log(1 - u)) \right) \quad (4)$$

For given patch sequences S , a patch selection mask $z \sim \text{Bern}(z | \theta)$ is sampled and the patch sequence S will be multiplied by this mask $S' = S * z$ to stochastically set the hidden embedding of patches to 0 at a rate of θ . Instead of setting θ as a fixed value, we made it a parameter to be optimized which depends on not only the patch status but also the overall status of the patch sequence. For a feature aggregator $f(x)$, given a mini-batch of size M , the loss objective

function to optimize the parameter θ can be written as (3), where z is a patch selection mask. The optimization of θ requires a backpropagation of the discrete variable z . We sample from concrete distributions with temperatures t that produce values between 0 and 1, rather than from discrete Bernoulli variables. The specific relaxation $\tilde{z}$ of the Bernoulli distribution $\text{Bern}(z | \theta)$ is shown in (4).

We have drawn Fig. 3 to depict a visual representation of our proposed attention mechanism. As shown in Fig. 3, the patch sequences were first fed into a Bi-LSTM network [36] for feature aggregation. The LSTM output was then passed through a linear layer to calculate the dropout probability for each patch. The probability was used as a patch sequence mask to mask the LSTM output vector. Finally, the masked LSTM was fed into a three-layer MLP network to obtain the Gleason score.

3.5 Model complexity

We briefly analyzed the model's complexity in terms of training efficiency and memory usage. Since the training of our model was based on frozen embeddings of patches, feature extraction using the self-supervised pre-trained ResNet50 is a one-time process. The main computational overheads are our sequence aggregator LSTM and the fully connected

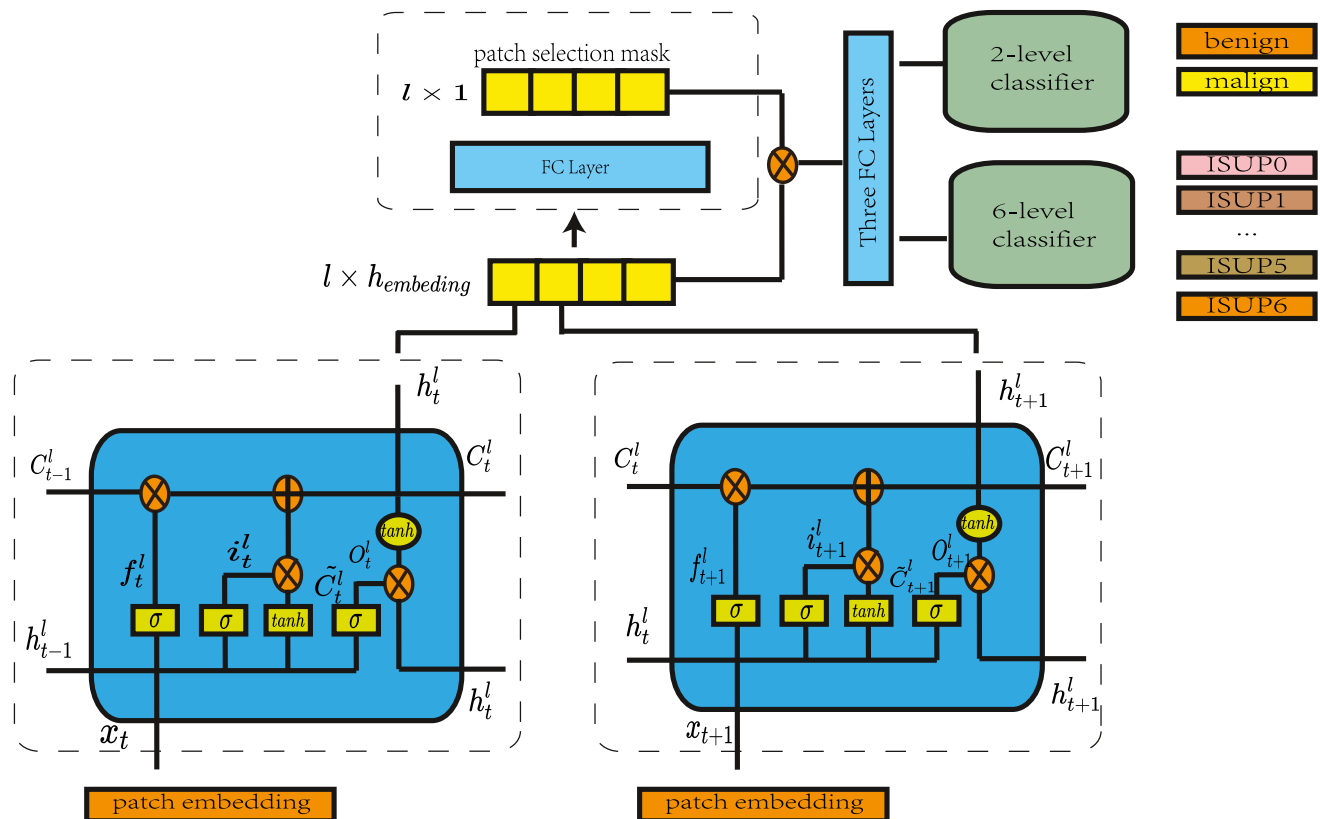


Fig. 3 Schematic of our proposed attention LSTM aggregator

Table 4 Model complexity of our method and other state-of-the-art methods

Method	Modules	Params	MACs
Attention-MIL [34]	Resnet50	25.56M	2.159T
	Attention module	1.089M	108.81M
CLAM-MB [26]	Resnet50	25.56M	2.159T
	Attention module	25.56M	108.81M
GNN [22]	Resnet50	25.56M	4.134T
	Graph convolution	200K	152M
	Attention-pooling	795K	871M
Our method	Resnet50	25.56M	0M
	Attention-LSTM	16.79M	1.68G

layers. The parameters and MACs of our pipeline, as well as other advanced methods for Gleason grading, are listed in Table 4. MACs are calculated by assuming that a slide image is divided into 100 patches, each measuring 512x512 pixels. The MACs value of our method is significantly smaller than that of other methods, such as GNN [22] which requires 4.135T MACs to process a slide image. In contrast, our method only requires 1.68G MACs to train a slide image. Our approach is also memory efficient because the memory requirement of our method is determined solely by the size of the cropped patch. Therefore, there won't be any memory bottleneck, regardless of the size of the pathology image being processed.

4 Experiment and result

4.1 Evaluation metric

To demonstrate the superiority of our approach, we use accuracy (ACC), F1-score (F1), Cohen's quadratic kappa, and linear kappa to evaluate the performance of our model.

4.2 Implementation details

We performed all image processing work and network training using a local computer with the hardware and software configurations listed in Table 5. We used Python 3.7.5 language and PyTorch 1.3.1 framework to implement our method. When training our model, we utilized the Adam optimizer with an initial learning rate set to $3e-4$ and adopted a cosine annealing decay schedule to update the learning rate of the optimizer. The maximum number of epochs was set to 100. After completing each training epoch, we calculated the kappa metric for the validation set. If the validation kappa metric reaches its best value, we will save the weights.

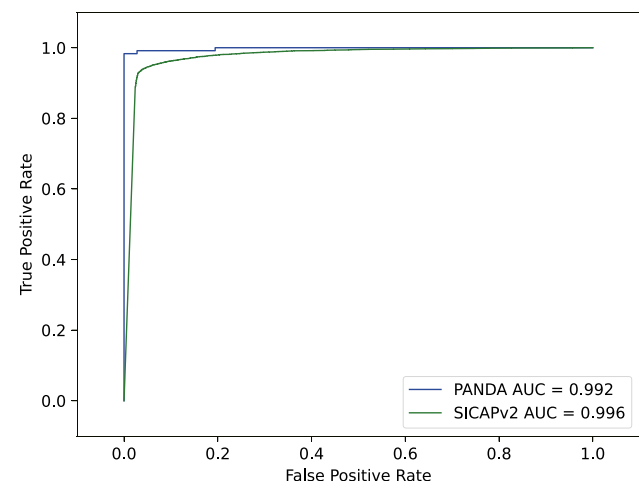
Table 5 Hardware and software framework

Name	Description
RAM	32GB
CPU	14 core AMD EPYC 7453
GPU	GeForce Rtx2080ti
OS	Ubuntu 18.04
Software Framework	python = 3.75, pytorch = 1.3.1, torchvision = 0.4.2, openslide = 3.4.1

4.3 Cancer diagnosis result

Prostate cancer detection is a binary classification task that aims to determine whether a Whole Slide Image (WSI) contains malignant tissue. We assessed the model's ability to detect cancer using metrics such as AUC, sensitivity, and specificity. As shown in Fig. 4, our method achieved an AUC value of 0.992 (95% CI 0.989-0.994) for 5-fold cross-validation on the PANDA dataset and an AUC value of 0.996 (95% CI 0.992-0.998) on the external SICAPv2 dataset. To provide a detailed representation of the probability distribution projected by our model using the PANDA and SICAPv2 datasets.

We also plotted histograms of the slide prediction probabilities for the two datasets. As shown in Fig. 5, the blue histograms represent the true cancer samples, while the green histograms represent the non-cancer samples. We used a threshold of 0.50 to determine whether the predicted result corresponds to cancerous or non-cancerous tissue. From Fig. 5, we can see that the system can detect the majority of the cancer samples with a high level of accuracy. On the PANDA and SICAPV2 datasets, our model achieved false negative rates of 1.6% and 0.65%, respectively. The lower

**Fig. 4** AUC for Prostate cancer detection with PANDA and SICAPv2 datasets

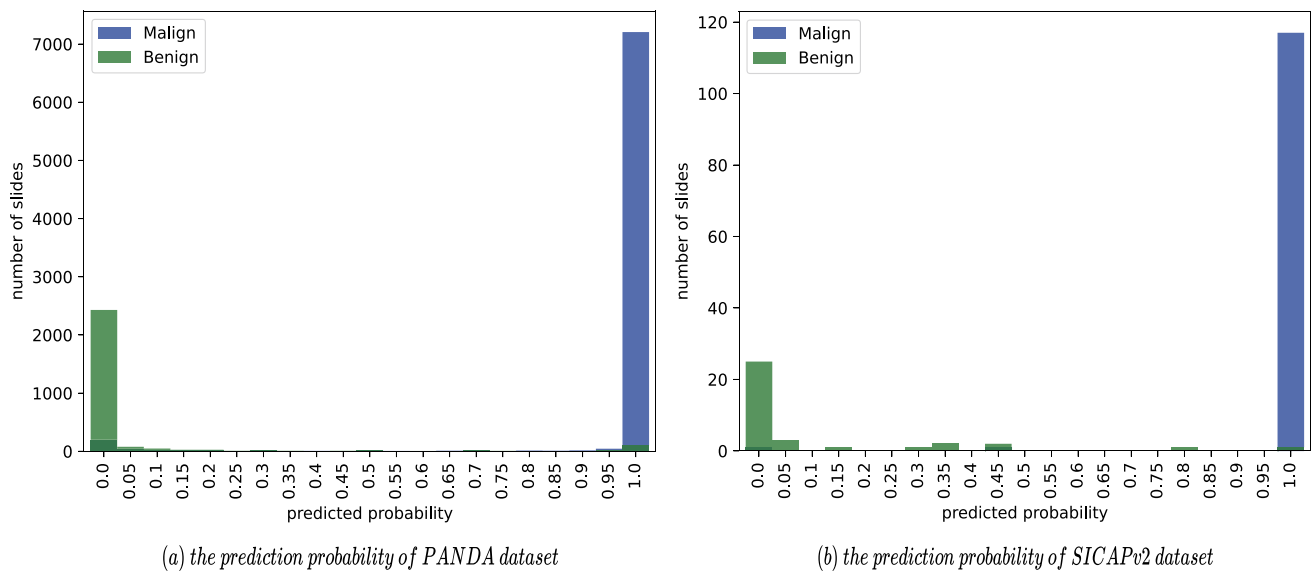


Fig. 5 (a), (b) are the histograms that show the predicted probability on the panda dataset and sicapv2 dataset

false negative rate means that our CAD-Assist systems would not overlook potential cancer cases.

4.4 Gleason grading result

Grading prostate cancer is a regression task that aims to classify Whole Slide images (WSI) into 6 levels of ISUP grade (ISUP 0, 1, 2, 3, 4, 5). We have listed the kappa, ACC, and F1 metrics of other advanced methods, as well as our method, for 5-fold cross-validation with the PANDA dataset in Table 6. As shown in Table 6, our model achieved an accuracy of 0.786 (95% CI: 0.776-0.791), a linear kappa of 0.849 (95% CI: 0.847-0.852), and a quadratic weighted kappa of 0.920 (95% CI: 0.914-0.922) for the PANDA dataset (n=10,615) using 5-fold cross-validation. The metrics of three early baseline methods, Attention-MIL [34], CLAM-MB [26], and DSMIL

[37], as well as the current state-of-the-art method GCN-MIL, are also shown in Table 6. Our approach goes beyond the previously proposed methods and has achieved comparable kappa and accuracy metrics to the best available method.

To validate the generalization of the proposed method, we used two separate datasets. The first dataset was the internal test dataset from the PANDA competition and another dataset utilized was the public dataset SCIAPv2. As can be seen in Table 7, for the SICAP dataset, the proposed method achieved excellent results with a kappa value of 0.8668 for the Gleason grading task, which is 4% better than the method proposed by Julio Silva-Rodriguez [12] 4%. The kappa value was slightly worse than that of Pushpak Pati's method [39], but our F1 score was much higher than theirs. For the PANDA internal test dataset, we achieved a quadratic kappa value of 0.930, which is comparable to the performance of the previous best

Table 6 Gleason Score metrics obtained by our method and other advanced methods using 5-fold Cross-validation with panda dataset

Method	Linear kappa	Accuracy (%)	Quadratic kappa
Silva-Rodríguez et al. [12]	—	—	0.8054
Attention-MIL [34]	0.611(0.591-0.625)	0.459 (0.444-0.483)	0.779 (0.758-0.783)
CLAM-MB [26]	0.611(0.589-0.626)	0.459(0.441-0.487)	0.780(0.760-0.792)
DSMIL [37]	0.601(0.585-0.615)	0.455(0.436-0.477)	0.769(0.758-0.788)
GCN-MIL1 [22]	0.736(0.731-0.738)	0.623(0.616-0.631)	0.855(0.849-0.859)
GCN-MIL2 [22]	0.812(0.807-0.816)	0.732(0.721 - 0.741)	0.897(0.887-0.905)
GCN-MIL3 [22]	0.850(0.847-0.854)	0.785(0.775 - 0.789)	0.919(0.911-0.924)
Singhai et al. [38]	—	0.831	0.930
Vision Transformer [9]	0.813(0.807 - 0.822)	0.726 (0.721 - 0.736)	0.897(0.882 - 0.902)
Swin Transformer [28]	0.832(0.827 - 0.838)	0.742 (0.737 - 0.750)	0.907(0.901 - 0.912)
Ours	0.849(0.847 - 0.852)	0.786 (0.776 - 0.791)	0.920(0.914 - 0.922)

Table 7 Gleason Score metrics obtained using our method and other advanced methods, employing a 5-fold cross-validation technique with the Panda dataset

Method	Panda internal dataset			
	Quadratic kappa	Linear kappa	Accuracy (%)	F1 score
Attention-MIL [34]	0.765	NA	NA	—
CLAM-MB [26]	0.781	NA	NA	—
DSMIL [37]	0.728	NA	NA	—
GCN-MIL-1 [22]	0.853	NA	NA	—
GCN-MIL-2 [22]	0.894	NA	NA	—
GCN-MIL-3 [22]	0.931	NA	NA	—
Ours	0.93	NA	NA	—
SICAP dataset				
Supervised [16]	0.817	—	—	—
Silva-Rodríguez et al. [12]	0.8299	—	—	—
Pushpak Pati et al. [39]	0.89	—	—	0.586
Ours	0.8668(0.8478-0.8729)	0.7756(0.775-0.789)	0.6912(0.6843-0.7011)	0.6745(0.6717-0.6758)

method. The possible explanation for the high performance achieved by our method is that the attention LSTM sequence aggregator can aggregate a broader range of features in the patch sequence and enable the sequence aggregator to automatically select the most suitable patch sequence.

4.5 Comparisons with other architecture

To demonstrate the progress of our proposed method, it is essential to compare it with the current state-of-the-art architectures in the field of vision. The Transformer architecture, which originated in the field of Natural Language Processing (NLP), has become the state-of-the-art approach for a growing number of vision tasks [9, 27, 28]. The main highlights of the ViT architectures are as follows: cutting the input image into small patches, converting these patches into token embeddings, and then feeding these token embeddings, along with position embeddings, into multi-headed self-attention modules. Swin Transformer is an enhanced version of ViT that substitutes the global attention mechanism of the original ViT model. It restricts the attention mechanism to a specific window location and utilizes a sliding window for information exchange between windows.

We incorporated this approach into our pipeline and substituted our feature extraction model with the ViT and Swin Transformer models. Due to the unavailability of pre-training weights specifically for pathology images, we used the pre-trained weights from ImageNet to extract patch features. The detailed results can be found in Table 6. From Table 6, we can see that utilizing both ViT and Swin Transformer as feature extractors did not result in outstanding performance for our model. The quadratic weighted kappa values were only

0.897 and 0.907, respectively, representing a performance degradation of approximately 2.6% and 1.4% compared to our pipeline that utilizes self-supervised learning pre-trained backbones (Fig. 6).

4.6 Comparison with other aggregators

Most weakly-supervised trained models for pathology tasks require aggregating patch-level information to generate slide-level labels. Various types of aggregators have been applied to pathology images thus far. Table 8 lists some of the commonly used aggregators and the corresponding experimental results obtained by replacing our aggregator with these alternatives. We utilized the same MLP layer as the ViT architecture, comprising two linear layers, to ensure fairness. Additionally, we set the maximum number of epochs to 30 in order to reduce training time.

From Table 8, we can see that max-pooling performed the best, and mean pooling performed the worst among the heuristic aggregators for the prostate cancer grading task. The performance of contact pooling, which combines mean pooling and maximum pooling features, improved by approximately 0.57%, 3.4%, and 1.7% for quadratic kappa, ACC, and linear kappa, respectively, compared to max pooling. This demonstrates that the mean statistic influences Gleason's grading. Attention-MIL [34] and CLAM-MB [34] are parameterized aggregators that incorporate the attention mechanism into the multi-instance learning paradigm. Their performance was approximately 3% better than contact pooling, and there is minimal difference in performance between these two attention-based multi-instance learning approaches. The ViT aggregator performed slightly worse

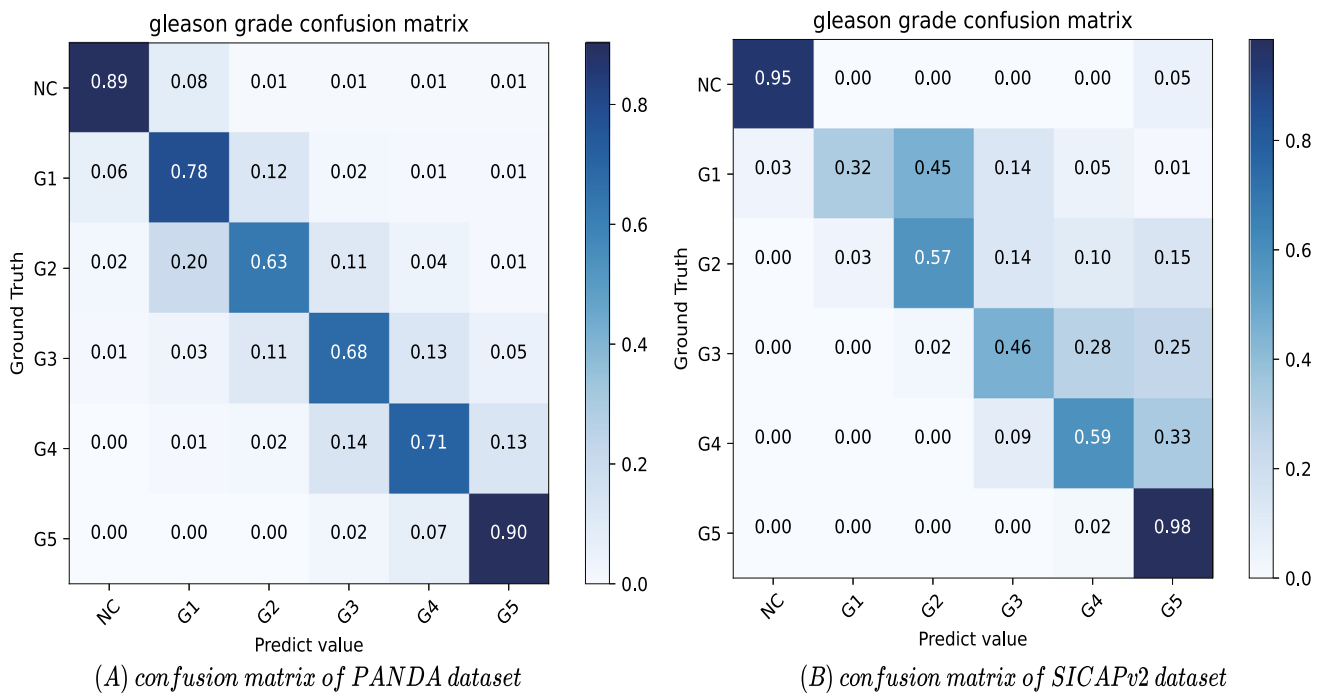


Fig. 6 ISUP grade confusion matrix between predictions and reference standard

than expected, possibly because VITs require a large number of samples to complete model training, and the size of our datasets does not allow the VIT to train adequately. Besides, Vits should globally fix the patch sequence length instead of dynamically padding the batches. This is an unfavorable factor that also affects their performance.

RNN and GRU are sequence-based aggregators, and in our scenario, GRU performs significantly better than RNN. From the table, we can see that our proposed aggregator achieved the best performance compared to other aggregators when conducting cross-validation on PANDA datasets. Our quadratic kappa, accuracy, and linear kappa values reached 0.903, 0.745, and 0.822, respectively, surpassing those of other feature aggregators.

4.7 Feature ranking

Truncating patch sequences can avoid introducing too much null information when aligning the patch sequence. Before truncating the sequence, it is necessary to rank patch sequences to make the truncating operations more likely happen on unimportant patches.

Traditional patch ranking methods, such as random forest, mutual information, and neural networks (NN), can handle feature ranking in a predetermined order. When the patch sequences are of variable length and have no fixed order, these traditional feature ranking methods will not work well. Attention-based ranking methods, such as Attention-MIL [34] and CLAM-MB [34] may be better choices for

Table 8 Performance of our feature aggregator and other types of aggregators

Aggregation	Evaluate metrics		
	Quadratic kappa	Accuracy	Linear kappa
Average-pooling [35]	0.823(0.822 - 0.824)	0.556(0.554 - 0.559)	0.682(0.681 - 0.684)
Max-pooling [35]	0.866(0.863 - 0.867)	0.642(0.641 - 0.644)	0.753(0.752 - 0.754)
Contact-pooling [35]	0.871(0.870 - 0.872)	0.664(0.662 - 0.665)	0.766(0.764 - 0.767)
GEM pooling [35]	0.859(0.858 - 0.861)	0.644(0.642 - 0.645)	0.748(0.747 - 0.751)
Attention-MIL [34]	0.883(0.881 - 0.884)	0.685(0.684 - 0.686)	0.783(0.780 - 0.784)
CLAM-MB [26]	0.869(0.869 - 0.872)	0.689(0.688 - 0.692)	0.783(0.781 - 0.784)
RNN [34]	0.857(0.856 - 0.859)	0.651(0.649 - 0.653)	0.753(0.752 - 0.757)
GRU [34]	0.870(0.869 - 0.881)	0.682(0.680 - 0.684)	0.775(0.772 - 0.779)
Vision transformer [9]	0.848(0.846 - 0.851)	0.616(0.614 - 0.619)	0.731(0.729 - 0.733)
Our aggregation	0.903(0.894 - 0.907)	0.745 (0.732 - 0.756)	0.822(0.817 - 0.829)

Table 9 Experimental results with TOP patch sequences ranked by different feature ranking methods

Ranking methods	Top 50 patches			Top 100 patches		
	Quad kappa	Acc	Linear kappa	Quad kappa	Acc	Linear kappa
Mutual info	0.8741	0.6507	0.7582	0.8833	0.6911	0.7876
Random forest	0.8899	0.6966	0.7927	0.8911	0.68753	0.7895
Random suffer	0.8882	0.7118	0.7976	0.8848	0.7078	0.7962
Attention-MIL [34]	0.8924	0.71476	0.8036	0.9044	0.7358	0.8232
CLAM-MB [26]	0.8962	0.6914	0.8232	0.9012	0.73244	0.8183
Our method	0.9159	0.7573	0.8394	0.9101	0.7578	0.8333

handling these ranking tasks. To demonstrate the effectiveness of our proposed attention mechanism for selecting patch sequences, we ranked the patch sequences according to our attention weights. We then selected the top-ranked N patch sequences and retrained the model. Metrics derived from cross-validation experiments on the PANDA dataset using the top N patch sequences ranked by our attention method and other methods are listed in Table 9. As shown in Table 9, our feature aggregator achieved a quadratic kappa value of 0.9159, an accuracy of 0.7573, and a linear kappa value of 0.8394 using only the top 50 ranked patch sequences. This performance far surpasses that of traditional methods for feature ordering and feature ranking that utilize attention. When using the top 100 patch sequences, the performance of our method degraded, but it still outperformed methods such as Attention-MIL [34], CLAM-MB [26].

5 Discussion

5.1 Performance analysis

For the computer-aided diagnosis of prostate cancer, CNNs need to handle multi-megapixel images that contain both global contextual and local textural information. Direct end-to-end training using these oversized images can encounter memory bottleneck issues due to the large intermediate activations that conventional backpropagation needs to store. To address the memory issues, we have implemented a two-step approach to handle the tasks of feature extraction and supervised classification. First, we divided the large pathological images into smaller patches. Then, we used feature extractors trained through self-supervised learning to extract representations from these patches. Through this approach, we can ensure that the memory requirements of our method depend solely on the size of the cropped patches. This allows us to completely eliminate memory bottlenecks, regardless of the size of the pathology image being processed. The computational complexity of our method is also low. The number of Multiply-Accumulate operations (MACs) required for our model to process a slide is 1.68G when it is cropped into

100 patches. This is significantly smaller than other methods, such as GNN [22] which requires 4.135T MACs to process a slide.

The feature aggregator is also a highlight of our work. Although there are already advanced feature aggregators such as GNN, RNN, and VIT, few works have recognized that pathological images often contain a significant amount of redundant information, and pathologists can score the entire Whole Slide Images (WSI) based on a small percentage of the slides. Mohsin et al. [40] used the top 1% or top k patches of the maximum activation as the fixed patch. They also selected other random combinations of patches to form the collections of patches. Experiments showed that their method of combining patches can effectively improve the overall classification performance. However, such a rule-based patch combination method can suffer from poor generalization in the field of pathological images. It is necessary to autonomously select the patch sequence based on the patch situation and the current state of the patch sequence. Our feature aggregator can meet this requirement by dynamically selecting patch sequences that are important for Gleason grading using variational dropout. You can think of our feature aggregator as a distinct type of attention aggregator, where the attention weight represents the probability of dropout in the patch sequence. This can serve as a more effective method for ranking features. Subsequent experiments have confirmed that feature ranking based on our attention mechanism worked better than other feature ranking methods, such as Attention-MIL [34] or CLAM-MB [26]. The quadratic kappa obtained from Gleason grading tasks using the top 50 patches selected by our ranking method is 2.6% higher than both Attention-MIL [34] and the CLAM-MB [26] ranking methods. Similar improvements can also be observed in other metrics, including accuracy and linear Kappa.

We have also conducted experiments to compare our methods with other advanced methods. Experiment results confirmed that the performance of our methods is comparable to advanced methods such as Vision Transformer [9], CLAM-MB [34] and GNN [22] in all metrics.

Our pipeline yielded an AUC ranging from 0.985 to 0.992 for cancer detection. Additionally, it achieved a quadratic

weighted kappa value of 0.931 and 0.810 for the internal and external test sets, respectively, for Gleason grading. This indicates a performance improvement of 4% compared to the previous best model. It is worth noting that the current state-of-the-art Vision Transformer does not perform well when used as a feature aggregator for pathology images. One possible explanation for this is that the cropped pathology patch set is not strictly sequential data, but rather order-independent aggregate data, and VITs require a large number of samples to complete model training, and the size of our datasets does not allow the VIT to train adequately. In addition, one of the challenges that Vit has to face is the variability of the patch sequence. When using VIT as a feature extractor, its performance is still poor primarily due to the absence of pre-trained weights on large-scale pathology images. Currently, we can only utilize weights trained on the ImageNet dataset, which is not sufficient for extracting features from pathology images. This phenomenon also highlights the significant limitations of our proposed approach. Our method heavily relies on the availability of self-supervised, pre-trained models. However,

training such a feature extractor is both costly and nearly impossible with only a few GPU graphics cards. Furthermore, the patch cropping scheme is a significant drawback of our method. It is crucial to maintain consistency between the cropping methods used for the training and test datasets to prevent a significant decline in performance.

5.2 Visualization analysis

In addition to analyzing the statistical performance of the proposed model's predictions, we also provide an intuitive analysis of our methods. Our attention LSTM model utilizes an attention module to redistribute attention and aggregate the patch sequence in a bag, ultimately making the final prediction. These attention scores can reveal the contribution and importance of each patch to the final output. In addition to patch attention, we also display segmentation masks, which are important criteria for determining the Gleason grade of the slide. From Fig. 7, we can see that the regions our attention module focuses on essentially match the segmentation



Fig. 7 Interpretability and visualization. left images are segmentation masks showing which parts of the images led to the ISUP grade. Right images are patch selection attention generated our method

mask. This indicates that our proposed attention method can effectively identify the regions that determine the score and enhance the performance of the slide-level Gleason grading model.

5.3 Limitations

Although cross-validation was used to improve the model's ability to generalize, we still noticed a decline in model performance in metrics such as kappa, f1, and accuracy when conducting independent tests on various datasets. The decline in predictive performance during external validation may be attributed to variations in data distribution. But also related to the image preprocessing strategy we used. Therefore, it is desirable to use a consistent cropping strategy for both the training dataset and the test dataset.

6 Conclusion

In summary, we have developed a novel slide-level predictor for Gleason grades in pathological images. This predictor has the capability to automate the Gleason scoring of prostate cancer. The adoption of self-supervised learning enables us to work with gigapixel histopathology images using only a single GPU graphics card to perform Gleason grading and cancer detection tasks. We have also developed a novel attention aggregator that can automatically select the optimal sequence of patches. To the best of our knowledge, we are pioneers in introducing the concept of patch selection to patch aggregators. This allows the Gleason scoring task to benefit from the diversity of data in the patch subsequences. Subsequent experiments demonstrate the effectiveness of our proposed feature aggregator compared to other patch sequence aggregators. Our method requires an additional step to crop the patch, which is time-consuming. In future work, we will improve the way we crop the patches to save preprocessing time. Additionally, we will test the applicability of our approach using other pathology datasets.

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Author Contributions Liang Ma and Hao chen performed the data analysis and wrote the manuscript. Liang Ma, Hao chen and Ming Gong conduct the experiment together.

Data Availability Data used in the present study are publicly available, and ethical approval and informed consent were obtained in each original study.

Code Availability On the <https://github.com/maliang07/mycodes.git>

Declarations

Conflicts of interest The authors declare that there is no conflict of interest.

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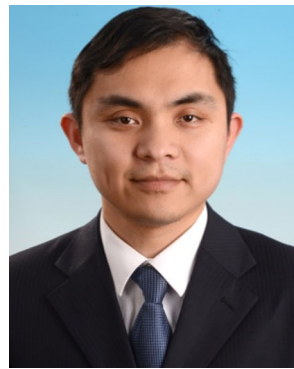
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